MB504P – METHODS IN MOLECULAR BIOLOGY

Practical Exam Notes

Prepared by: *Sir Mughal*

Main Group link: https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN

Subject Group link: https://chat.whatsapp.com/GVlgfk6B5tHDSsgXbvYMV8

Compiled by SIR MUGHAL

M.C.Qs (Multiple Choice Questions)

1. DNA precipitation is used for **PRECIPITATION OF DNA**.
2. Washing of RNA is done by **ETHANOL**.
3. Color of Ethidium Bromide when observed under UV light is **ORANGE**.
4. Who discovered non virulent streptococcus pneumoniae? **FREDRICK GRIFFITH**
5. What is primary use of SDS-PAGE? **ANALYSIS OF PROTEIN**
6. In which step of PCR, DNA is separated by heating? **DENATURATION**
7. During DNA extraction, **PROTEINASE K** is used for protein digestion.
8. What is the ration of PCI (Phenyl Chloroform Isoamyl Alcohol)? **25:24:1**
9. DNA concentration is estimated by measuring absorbance at which wavelength? **260 nm**
10. In gel electrophoresis, **SMALL** size DNA molecules move faster than **LARGE** ones.
11. What is the purpose of phase separation in organic DNA extraction? **PURIFICATION**
12. RFLP stands for **RESTRICTION FRAGMENT LENGTH POLYMORPHISM**.
13. Which process follows bacterial transformation to select for cells that have taken up foreign DNA? **SELECTION**
14. By which method, field blood sample must be stored during sampling? **COLD CHAIN**
15. The temperature of incubation during digestion step should be **56°C.**
16. In which layer RNA residues during phase separation using Trizol? **AQUEOUS**
17. In a PCR mixture, **PRIMERS** ensures PCR produced size specificity.
18. Which enzyme is commonly used in RFLP? **RESTRICTION ENDONUCLEASE**
19. What is the purpose of standard DNA ladder in gel electrophoresis? **SIZE ESTIMATION**
20. In which step of PCR separation of two strands of DNA occurs? **DENATURATION**
21. Bacterial transformation is **HORIZONTAL TRANSFER**.
22. Why EDTA buffer is used in lysis of Red Blood Cells? **CHELATION**
23. Which one is not used for DNA extraction? **RBCs**
24. At what temperature, Ethidium Bromide is added to agarose gel? **55°C**
25. First step of DNA extraction is **LYSIS**.

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

SHORT QUESTIONS

1. Write principles of PCR.

Principles of PCR

- ✓ ***Template DNA:*** The DNA containing the target sequence to be amplified.
- ✓ ***Primers:*** Short single-stranded DNA sequences that bind to the target DNA and provide a starting point for DNA synthesis.
- ✓ ***DNA Polymerase:*** A heat-stable enzyme (commonly Taq polymerase) that synthesizes new DNA strands.
- ✓ ***Nucleotides (dNTPs):*** The building blocks (A, T, G, and C) used to synthesize new DNA.
- ✓ ***Thermal Cycling:*** PCR consists of three repeated steps:
  - ***Denaturation (94–98°C):*** Double-stranded DNA separates into single strands.
  - ***Annealing (50–65°C):*** Primers bind (anneal) to complementary sequences on the template DNA.
  - ***Extension (72°C):*** Taq polymerase extends the primers by adding nucleotides, forming new DNA strands.
- ✓ ***Exponential Amplification:*** Each cycle doubles the amount of target DNA, resulting in millions of copies after 25–35 cycles.

2. Why nuclease free water is used for DNA storage after extraction of DNA?

Nuclease-free water is used for DNA storage after extraction because it is free from enzymes such as DNases and RNases that can degrade DNA. It prevents DNA fragmentation and maintains the quality, purity, and stability of the extracted DNA. Storing DNA in nuclease-free water ensures that the DNA remains intact and suitable for further molecular techniques such as PCR, sequencing, and other genetic analyses.

3. How competent cells are formed chemically?

Competent cells are formed chemically by treating bacterial cells with divalent cations such as calcium chloride ($CaCl_2$). This treatment neutralizes the negative charges on both the bacterial cell membrane and DNA, reducing repulsion and allowing DNA to come closer to the cell surface. The cells are then subjected to heat shock, which temporarily increases membrane permeability and creates pores through which foreign DNA can enter. After DNA uptake, the cells are allowed to recover and can express the introduced genetic material. This method is commonly used for bacterial transformation.

4. In DNA extraction, after the phase separation, three distinct layers are formed. Which layer has DNA?

After phase separation in DNA extraction, three distinct layers are formed: the upper aqueous layer, middle interphase, and lower organic layer. The DNA is mainly present in the upper aqueous layer because DNA is soluble in water. The aqueous layer contains nucleic acids, while proteins and lipids are separated into the interphase and organic layer, respectively. The DNA is then precipitated from the aqueous phase using alcohol such as ethanol or isopropanol for further purification.

Compiled by SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

5. Write characteristic that should be considered while designing a primer.

The following characteristics should be considered while designing a primer:

- ✓ ***Length:*** Primers should usually be 18–25 nucleotides long.
- ✓ ***GC content:*** Should be around 40–60% for proper binding and stability.
- ✓ ***Melting temperature (Tm):*** Forward and reverse primers should have similar Tm, generally around 50–65°C.
- ✓ ***Specificity:*** Primers should bind specifically to the target DNA sequence and avoid non-specific amplification.
- ✓ ***Avoid secondary structures:*** Primers should not form hairpins or primer-dimers.
- ✓ ***3′ end stability:*** The 3′ end should allow efficient extension by DNA polymerase.

6. Define stock solution and working solution.

Stock solution: A stock solution is a concentrated solution of a reagent prepared with a precisely known concentration. It is usually prepared in larger amounts, stored under suitable conditions, and used as a source to prepare diluted solutions. Stock solutions help maintain accuracy and consistency in laboratory experiments.

Working solution: A working solution is a diluted form of a stock solution that is prepared at the required concentration for immediate use in an experiment. It contains the reagent concentration suitable for a specific procedure and reduces the need to prepare fresh solutions repeatedly from the original chemical.

7. Write a short note on Multiplex PCR.

Multiplex-PCR

Multiplex PCR is an advanced form of Polymerase Chain Reaction in which two or more pairs of primers are used in a single PCR reaction to amplify multiple DNA targets simultaneously. It allows the detection of several DNA sequences in one reaction, making the process faster, more efficient, and cost-effective compared to performing separate PCR reactions. Successful multiplex PCR requires careful selection of primers with compatible melting temperatures (Tm), specificity, and minimal interaction between primer pairs. The amplified products are usually separated and identified by agarose gel electrophoresis. Multiplex PCR is commonly used in clinical diagnosis, detection of infectious agents, and identification of genetic mutations, forensic analysis, and molecular biology research.

8. How does SDS aids in the removal of proteins from DNA during DNA extraction?

SDS (sodium dodecyl sulfate) aids in the removal of proteins during DNA extraction by acting as a detergent. It disrupts the cell membrane and nuclear membrane, releasing DNA and cellular components. SDS also denatures proteins by breaking their bonds and causing them to lose their structure. The denatured proteins can then be separated from DNA through processes such as centrifugation or organic extraction. This helps obtain pure DNA free from protein contamination for further molecular applications.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

9. What is the function of chloroform in RNA extraction?

Chloroform is used in RNA extraction to **sep**arate RNA from proteins, DNA, and other cellular components. It helps in phase separation when added with phenol or lysis buffer, forming an aqueous phase, interphase, and organic phase after centrifugation. RNA remains in the upper aqueous phase, while proteins and lipids move to the organic phase or interphase. This step helps obtain pure RNA with reduced contamination for downstream applications such as RT-PCR and gene expression studies.

10. What is the length of primer in a mini primer and conventional PCR? Also write the purpose of primer in a PCR.

PRIMER-LENGTH

In conventional PCR, primers are usually designed with a length of 18–25 nucleotides, which provides sufficient specificity and stable binding to the target DNA sequence. In mini primer PCR, shorter primers of about 10–15 nucleotides are used, allowing amplification of specific regions with shorter sequences.

Purpose of primer in PCR

Primers are essential components of PCR because they provide a starting point for DNA synthesis by DNA polymerase. They bind to complementary sequences on the template DNA during the annealing step and define the specific region of DNA that will be amplified. A pair of primers (forward and reverse) is used to amplify the target DNA segment, ensuring accurate and specific production of DNA copies during PCR cycles.

11. What is the purpose of lysis buffer?

Lysis buffer is used to break open cells and release cellular components, including DNA, RNA, and proteins, during nucleic acid extraction. It disrupts the cell membrane and nuclear membrane with the help of detergents and salts, allowing the release of genetic material. It also helps protect nucleic acids from degradation by inhibiting nucleases and maintains a suitable chemical environment for efficient extraction.

12. Why we use nuclease free water for the blank purpose in Nano drop?

Nuclease-free water is used as a blank in NanoDrop because it does not contain DNA, RNA, or nucleases that can interfere with absorbance measurements. It provides a baseline reading by measuring the background absorbance of the solvent, allowing the instrument to accurately calculate the concentration and purity of the nucleic acid sample. This helps prevent false readings and ensures reliable DNA/RNA quantification.

13. What is the function of $CaCl_2$ in the preparation of chemically competent cells?

$CaCl_2$ (calcium chloride) plays an important role in the preparation of chemically competent cells by increasing the ability of bacterial cells to take up foreign DNA. The calcium ions neutralize the negative charges on the bacterial cell membrane and DNA, reducing repulsion between them and allowing DNA to attach to the cell surface. $CaCl_2$ also helps stabilize the cell membrane and, along with heat shock treatment, facilitates the entry of plasmid DNA into the cells during transformation.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

14. What is molal and molar solution?

Molar solution (Molarity):

A molar solution is a solution that contains one mole of solute dissolved in one litre of solution. Molarity is expressed as mol/L (M) and depends on the volume of the final solution. It is commonly used for preparing reagents and solutions in laboratory experiments.

Molal solution (Molality):

A molal solution is a solution that contains one mole of solute dissolved in one kilogram of solvent. Molality is expressed as mol/kg (m) and depends on the mass of the solvent, making it independent of temperature changes. It is mainly used in studies involving colligative properties of solutions.

15. What is the purpose of restriction enzyme mapping?

Restriction enzyme mapping is used to determine the location and order of restriction enzyme recognition sites within a DNA molecule. It involves cutting DNA with specific restriction enzymes and analyzing the resulting fragments to create a physical map of the DNA. This technique helps in identifying DNA fragments, confirming recombinant DNA constructs, studying gene organization, and analyzing plasmids or cloned DNA sequences.

16. What is the range of GC content?

The optimal GC content range for primers is 40–60%. This range provides proper binding stability between the primer and template DNA. A very low GC content may result in weak binding, while a very high GC content can cause strong binding and formation of secondary structures, affecting PCR efficiency.

17. Write precautions to use any harmful chemical.

Precautions while using harmful chemicals include:

- ✓ Always wear proper personal protective equipment (PPE) such as gloves, lab coat, and safety goggles.
- ✓ Handle chemicals in a well-ventilated area or fume hood to avoid inhalation of toxic vapors.
- ✓ Read the safety data sheet (SDS) and chemical labels before use.
- ✓ Avoid direct contact with chemicals and never taste or smell them directly.
- ✓ Use proper containers and labels for storage and disposal of chemicals.
- ✓ Dispose of chemical waste according to laboratory safety guidelines.
- ✓ Wash hands thoroughly after handling chemicals and clean the working area after use.

18. Write applications of RFLP.

Applications of RFLP (Restriction Fragment Length Polymorphism):

RFLP is a molecular technique used to detect variations in DNA sequences based on differences in restriction enzyme cutting patterns. Its applications include:

- ✓ Genetic disease diagnosis by identifying mutations or polymorphisms associated with inherited disorders.
- ✓ DNA fingerprinting and forensic analysis for individual identification.
- ✓ Paternity and relationship testing by comparing DNA patterns.
- ✓ Genetic mapping and studying the location of genes on chromosomes.
- ✓ Population genetics and evolutionary studies to analyze genetic variation among individuals or species.
- ✓ Identification of pathogens and strain differentiation in microbiological studies.

Compiled by SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

19. What is meant by 260/280 ration is Nano drop?

The 260/280 ratio in NanoDrop is a measure of the purity of DNA or RNA samples by comparing the absorbance at 260 nm and 280 nm wavelengths. Nucleic acids absorb light at 260 nm, while proteins (especially aromatic amino acids) absorb at 280 nm. For pure DNA, the ideal 260/280 ratio is approximately 1.8, while for pure RNA, it is around 2.0. A lower ratio indicates protein or phenol contamination, whereas a higher ratio may indicate contamination by other substances such as RNA or reagents. This ratio helps assess the quality of nucleic acid samples before further experiments.

20. How will you prepare 1M Tris HCl in a 0.5L solution? (M.W = 121.14 g/mole)

To prepare 1 M Tris-HCl in 0.5 L solution:

Formula: Weight (g) = Molarity (M) × Volume (L) × Molecular weight (g/mol)
= 1 × 0.5 × 121.14
= 60.57 g
Therefore, 60.57 g of Tris-HCl is required.

Preparation:

Dissolve 60.57 g of Tris-HCl in a suitable amount of distilled water. Adjust the volume to 500 mL (0.5 L) using distilled water. Mix thoroughly and store the prepared 1 M Tris-HCl solution under appropriate conditions.

21. Name reagents used in isolation of bacterial plasmid.

The reagents commonly used for bacterial plasmid isolation include:

- ✓ ***Tris-HCl buffer:*** Maintains a stable pH during extraction.
- ✓ ***EDTA:*** Chelates metal ions and inhibits DNases that can degrade DNA.
- ✓ ***SDS (Sodium dodecyl sulfate):*** Disrupts the cell membrane and denatures proteins.
- ✓ ***NaOH:*** Helps in cell lysis and denaturation of DNA and proteins.
- ✓ ***Potassium acetate:*** Neutralizes the solution and helps precipitate proteins and genomic DNA.
- ✓ ***RNase:*** Removes contaminating RNA from the plasmid preparation.
- ✓ ***Ethanol or isopropanol:*** Used to precipitate plasmid DNA.
- ✓ ***Nuclease-free water or TE buffer:*** Used to dissolve and store purified plasmid DNA.

22. Write methods of quantification of DNA.

Methods of DNA quantification:

- ✓ ***UV Spectrophotometry (NanoDrop):*** Measures DNA concentration by detecting absorbance at 260 nm. The purity of DNA is assessed using the 260/280 ratio.
- ✓ ***Fluorescence-based methods:*** Use fluorescent dyes that specifically bind to DNA, such as PicoGreen, to measure DNA concentration with high sensitivity.
- ✓ ***Agarose gel electrophoresis:*** DNA quantity is estimated by comparing the intensity of DNA bands with a known DNA ladder or standard.
- ✓ ***Real-time PCR (qPCR):*** Quantifies DNA by measuring amplification of the target DNA sequence during PCR cycles.
- ✓ ***Dye-based assays:*** Use DNA-binding dyes to estimate DNA concentration and are useful for samples with low DNA amounts.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

LONG QUESTIONS

1. Write ingredients and thermocyclic steps of conventional PCR.

Ingredients of Conventional PCR:

- ✓ ***Template DNA:*** The DNA sample containing the target sequence that needs to be amplified.
- ✓ ***Primers:*** Short single-stranded DNA sequences (forward and reverse primers) that bind to the complementary regions of the template DNA and provide a starting point for DNA synthesis.
- ✓ ***Taq DNA Polymerase:*** A thermostable DNA polymerase enzyme obtained from *Thermus aquaticus*. It synthesizes new DNA strands by adding nucleotides during the extension step.
- ✓ ***dNTPs (Deoxynucleotide triphosphates):*** The building blocks of DNA synthesis, including dATP, dTTP, dGTP, and dCTP.
- ✓ ***PCR Buffer:*** Provides the optimal pH and ionic conditions required for DNA polymerase activity.
- ✓ ***$MgCl_2$ (Magnesium chloride):*** Acts as a cofactor for Taq polymerase and is essential for efficient DNA amplification.
- ✓ ***Nuclease-free water:*** Used to adjust the final reaction volume and prevent DNA degradation.

Thermocyclic Steps of Conventional PCR:

PCR involves repeated cycles of three main temperature-dependent steps:

- ✓ ***Initial Denaturation (94–98°C):*** The double-stranded DNA template is heated to separate it into two single strands. This step ensures complete melting of the DNA before amplification begins.
- ✓ ***Denaturation (94–98°C):*** During each cycle, the double-stranded DNA is separated into single strands by breaking hydrogen bonds between complementary bases.
- ✓ ***Annealing (50–65°C):*** The temperature is lowered to allow primers to bind to their complementary sequences on the template DNA. The annealing temperature depends on the melting temperature (Tm) of the primers.
- ✓ ***Extension/Elongation (72°C):*** Taq DNA polymerase extends the primers by adding dNTPs in the 5′ to 3′ direction. New complementary DNA strands are synthesized.
- ✓ ***Final Extension (72°C):*** A final extension step allows complete synthesis of all DNA strands and ensures that any incomplete products are fully extended.
- ✓ ***Hold (4°C):*** The amplified DNA is maintained at a low temperature until further analysis.

2. What is bacterial transformation? Write down its principle.

Bacterial Transformation:

Bacterial transformation is a process in which foreign DNA (usually plasmid DNA) is introduced into a bacterial cell, resulting in the acquisition of new genetic characteristics. The bacterial cells that can take up external DNA are called competent cells**.** After transformation, the introduced DNA can replicate and express the desired gene inside the host cell.

Principle of Bacterial Transformation:

The principle of bacterial transformation is based on the ability of competent bacterial cells to take up extracellular DNA from their surroundings. Chemical treatment, such as calcium chloride ($CaCl_2$)**,** reduces the repulsion between negatively charged DNA and the bacterial cell membrane, allowing DNA to attach to the cell surface. A heat shock treatment creates temporary pores in the membrane, facilitating the entry of plasmid DNA into the cell. The transformed bacteria are then selected using appropriate markers, such as antibiotic resistance genes.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

3. Define steps involved in DNA extraction.

DNA extraction is a process used to isolate pure DNA from cells by removing other cellular components. The major steps involved are:

1. Cell Lysis

Cells are broken open using a lysis buffer containing detergents (e.g., SDS) to disrupt the cell membrane and nuclear membrane. This releases DNA along with other cellular components.

2. Removal of Proteins and Contaminants

Proteins and other impurities are removed using proteases, organic solvents, or salt treatment. EDTA is often used to inhibit nucleases that can degrade DNA.

3. Phase Separation

Organic extraction using phenol/chloroform separates cellular components into different phases. DNA remains mainly in the upper aqueous phase, while proteins and lipids are removed.

4. DNA Precipitation

DNA is precipitated from the aqueous phase by adding cold ethanol or isopropanol in the presence of salts. DNA appears as visible strands or a pellet after centrifugation.

5. DNA Washing and Drying

The DNA pellet is washed with 70% ethanol to remove impurities and excess salts. It is then dried to remove residual ethanol.

6. DNA Dissolution and Storage

Purified DNA is dissolved in nuclease-free water or TE buffer. The DNA is stored under suitable conditions for further applications such as PCR, sequencing, and cloning.

4. Briefly describe the troubleshooting during the isolation of bacterial plasmid.

Troubleshooting During Isolation of Bacterial Plasmid

Plasmid isolation may sometimes result in low yield, poor purity, or degraded DNA due to problems during different steps of the procedure. Common troubleshooting measures include:

1. Low plasmid DNA yield

- Use a fresh and healthy bacterial culture.
- Ensure proper bacterial growth conditions and use an appropriate amount of starting culture.
- Check that the lysis step is performed properly to release plasmid DNA.
- Ensure complete resuspension of the bacterial pellet before lysis.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

2. Contamination with genomic DNA

- Avoid vigorous mixing during the lysis step, as it can shear chromosomal DNA.
- Do not overuse the lysis solution.
- Carefully collect the clear supernatant after centrifugation.

3. Protein contamination

- Ensure proper removal of protein during purification.
- Use correct amounts of neutralization solution and perform proper centrifugation.
- A low 260/280 ratio indicates protein contamination.

4. RNA contamination

- Add RNase enzyme during plasmid isolation to remove unwanted RNA.
- Ensure RNase is active and properly stored.

5. Degraded plasmid DNA

- Use sterile, nuclease-free reagents and equipment.
- Avoid repeated freeze-thaw cycles.
- Store purified plasmid DNA at appropriate temperatures.

6. Poor DNA purity

- Wash the DNA pellet properly with ethanol.
- Remove residual ethanol before dissolving DNA.
- Use fresh buffers and maintain correct pH conditions.

5. Write working principle of Nano drop PCR.

Working Principle of NanoDrop

NanoDrop is a spectrophotometric technique used for the quantification and purity analysis of nucleic acids (DNA/RNA) and proteins. It works on the principle of UV-visible light absorbance, where nucleic acids absorb ultraviolet light at 260 nm and proteins absorb at 280 nm.

A small volume of DNA or RNA sample is placed on the measurement pedestal, and the instrument passes UV light through the sample. The amount of light absorbed by the sample is measured and used to calculate the concentration of nucleic acid according to the Beer–Lambert law. The 260/280 ratio is used to determine sample purity, where a ratio of approximately 1.8 indicates pure DNA and 2.0 indicates pure RNA.

NanoDrop also measures absorbance at 230 nm to detect contamination from substances such as salts, phenol, or other reagents. It provides rapid and accurate estimation of nucleic acid quantity and quality before applications such as PCR, sequencing, and cloning.

Compiled by SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

6. What is the purpose of lysis buffer in used in DNA isolation?

Purpose of Lysis Buffer in DNA Isolation

Lysis buffer plays an important role in DNA isolation by breaking open cells and releasing genetic material from the nucleus. It disrupts the cell membrane and nuclear membrane through the action of detergents such as SDS (sodium dodecyl sulfate), allowing DNA and other cellular components to be released into the solution.

Lysis buffer also contains components like EDTA, which binds divalent metal ions and inhibits nucleases (DNases) that can degrade DNA. Salts present in the buffer help maintain proper ionic conditions and assist in the separation of DNA from proteins and other cellular debris. Some lysis buffers also contain enzymes such as proteinase K to digest proteins and improve DNA purity.

Thus, lysis buffer ensures efficient cell disruption, protects DNA from degradation, and helps obtain pure, intact DNA suitable for downstream applications such as PCR, cloning, and sequencing.

7. Differentiate between RT PCR and Real Time PCR.

Difference Between RT-PCR and Real-Time PCR

RT-PCR (Reverse Transcription PCR)	Real-Time PCR (qPCR)
RT-PCR is a technique used to amplify RNA molecules by first converting RNA into complementary DNA (cDNA) using the enzyme reverse transcriptase.	Real-Time PCR is a technique used to monitor DNA amplification during each PCR cycle using fluorescent dyes or probes.
The main purpose is to detect and study RNA expression or RNA viruses.	The main purpose is to quantify the amount of DNA present in a sample.
It involves two steps: reverse transcription (RNA → cDNA) followed by PCR amplification.	It involves PCR amplification and detection of products simultaneously in real time.
The final product is usually analyzed after PCR completion using methods like gel electrophoresis.	The amplified DNA is measured during the PCR process using fluorescence signals.
It is commonly used for gene expression analysis and detection of RNA viruses.	It is commonly used for quantitative analysis of genes, pathogen detection, and measuring DNA copy number.

8. What are essential steps of RFLP?

Essential Steps of RFLP (Restriction Fragment Length Polymorphism)

RFLP is a molecular technique used to detect variations in DNA sequences based on differences in restriction enzyme cutting patterns. The essential steps involved are:

1. DNA Isolation

High-quality genomic DNA is extracted from the biological sample. The purified DNA is required for digestion with restriction enzymes.

Compiled by: SIR MUGHAL

2. Restriction Enzyme Digestion

DNA is treated with specific restriction enzymes that recognize and cut DNA at specific sequences. Differences in DNA sequences may create or remove restriction sites, producing fragments of different lengths.

3. Separation of DNA Fragments

The restriction fragments are separated according to their size using agarose gel electrophoresis. Smaller fragments migrate faster than larger fragments through the gel.

4. DNA Transfer (Southern Blotting)

Separated DNA fragments are transferred from the gel onto a membrane for detection. 5. Hybridization with DNA Probe. A labeled DNA probe is added that binds to the specific target DNA sequence. This allows identification of specific restriction fragments.

6. Detection and Analysis

The labeled fragments are visualized, and the banding pattern is analyzed. Differences in fragment sizes indicate genetic variation between samples.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

LMS Handling

by Sir Mughal

Reliable • Professional • Result-Oriented



Our Key Features

- Timely Activities
- Expert Team Members
- Maximum Quiz Scores
- Assignment Solutions by *Subject-Oriented Experts*
- Paid Project Handling
- Guaranteed Results
- Market Competitive Rates

Contact Us

0313-2310940

shahzeb.zoologist04@gmail.com